Jungwoo Park

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LinkedIn | GitHub | Google Scholar | Kaggle

EDUCATION

Korea University — *Seoul, South Korea*
Ph.D. in Computer Science and Engineering — *Sep. 2024 – Present*
- Advisor: Prof. Jaewoo Kang

Kwangwoon University — *Seoul, South Korea*
B.S. in Information Convergence — *Mar. 2018 – Feb. 2024*
- Overall GPA of 4.17/4.5, Major GPA of 4.43/4.5
- Full-ride merit scholarships × 2, one-quarter tuition scholarships × 3

RESEARCH INTERESTS

LLMs, Pre-Training, Mechanistic Interpretability

SELECTED PUBLICATIONS

1. **Jungwoo Park**, Taewhoo Lee, Jaewoo Kang†, "UltraBERT: Beyond the Limit of BERT Pre-Training," *Preprint*
2. **Jungwoo Park**, Jaewoo Kang†, "UltraBERT-Embed: A Compact Long-Context Encoder for Text and Code Embeddings," *Under review*
3. **Jungwoo Park***, Young Jin Ahn*, Kee-Eung Kim†, Jaewoo Kang†, "Monet: Mixture of Monosemantic Experts for Transformers," *ICLR 2025*
4. **Jungwoo Park**, Taewhoo Lee, Chanwoong Yoon, Hyeon Hwang, Jaewoo Kang†, "Outlier-Safe Pre-Training for Robust 4-Bit Quantization of Large Language Models," *ACL 2025*
5. **Jungwoo Park**, Hyeongsoon Hwang, Sanghoon Lee†, Jaewoo Kang†, "Meerkat2: Bridging Global Reasoning and Local Knowledge for Efficient Medical AI," *EMNLP 2026*
6. Jaehoon Yun*, Jiwoong Sohn*, **Jungwoo Park***, Hyunjae Kim, Xiangru Tang, Yanjun Shao, Yonghoe Koo, Minhyeok Ko, Qingyu Chen, Mark Gerstein, Michael Moor†, Jaewoo Kang†, "Med-PRM: Medical Reasoning Models with Stepwise, Guideline-verified Process Rewards," *EMNLP 2025*
7. Youngjin Ahn*, **Jungwoo Park***, Sangha Park, Jonghyun Choi†, Kee-Eung Kim†, "SyncVSR: Data-Efficient Visual Speech Recognition with End-to-End Crossmodal Audio Token Synchronization," *Interspeech 2024*

* Equal contribution. † Corresponding authors.

OTHER PUBLICATIONS

1. Suhyeong Park, Junha Jung, **Jungwoo Park**, Jaewoo Kang†, "Do All Visual Tokens Matter Equally? Object-Evidence Preserving Token Merging for Vision-Language Retrieval," *Preprint*
2. Taewhoo Lee, Minju Song, **Jungwoo Park**, Chanwoong Yoon, Junha Jung, Hyeon Hwang, Sanghoon Lee, Minbyul Jeong, Jaewoo Kang†, "Self-Assessment Reinforcement Learning for Search-State Awareness in Retrieval-Augmented Large Language Models," *Under review*
3. Sihyeon Park, **Jungwoo Park**, Hyunjae Kim, Jaewoo Kang†, Bumsoo Kim†, "Marginal Constrained Morphological Prototype Learning for Patch Search in Whole Slide Images," *MICCAI 2026*
4. Hyunjae Kim*, Xuguang Ai*, Sahana Srinivasan*, Aidan Gilson, Maxwell B. Singer, Krithi Pushpanathan, Qianqian Xie, **Jungwoo Park**, Serina Applebaum, Gabriel Dawei Yang, Minjie Zou, David Ziyou Chen, Ke Zou, Soshian Sarrafpour, Ji Liu, Yu Yin, Jimin Huang, Quang Ngoc Nguyen, Erping Long, Peixing Wan, Dianbo Liu, Richard Hintz, W. Jim Zheng, Sophia Y. Wang, Lucila Ohno-Machado, Hua Xu, Ron A. Adelman, Luciano V. Del Priore†, Yih-Chung Tham†, Qingyu Chen†, "LEME: Open Large Language Models for Ophthalmology with Advanced Reasoning and Clinical Validation," *npj Digital Medicine (2026)*
5. Yein Park, **Jungwoo Park**, Jaewoo Kang†, "ASGuard: Activation-Scaling Guard to Mitigate Targeted Jailbreaking Attack," *ICLR 2026*
6. Taewhoo Lee, Minju Song, Chanwoong Yoon, **Jungwoo Park**, Jaewoo Kang†, "The Curious Case of Analogies: Investigating Analogical Reasoning in Large Language Models," *AAAI 2026*
7. Hajung Kim*, Hoonick Lee*, Yewon Cho*, **Jungwoo Park***, Jueon Park*, Soyon Park*, Yan Ting Chok*, Seungheun Baek*, Donghyeon Lee*, Jaewoo Kang†, "Prompting Matters: Snippet-Aware Strategies for Biomedical QA with LLMs in BioASQ 13b," *CLEF 2025*
8. Yein Park, Chanwoong Yoon, **Jungwoo Park**, Minbyul Jeong, Jaewoo Kang†, "Does Time Have Its

Place? Temporal Heads: Where Language Models Recall Time-specific Information," *ACL 2025*

9. Hyunjae Kim, Hyeon Hwang, Jiwoo Lee, Sihyeon Park, Dain Kim, Taewhoo Lee, Chanwoong Yoon, Jiwoong Sohn, **Jungwoo Park**, Olga Reykhart, Thomas Fetherston, Donghee Choi, Soo Heon Kwak, Qingyu Chen, Jaewoo Kang†, "Small language models learn enhanced reasoning skills from medical textbooks," *npj Digital Medicine 8.1 (2025): 240*
10. Yein Park, Chanwoong Yoon, **Jungwoo Park**, Donghyeon Lee, Minbyul Jeong, Jaewoo Kang†, "ChroKnowledge: Unveiling Chronological Knowledge of Language Models in Multiple Domains," *ICLR 2025*

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PROJECTS

Jax/Flax Re-implementation of DeiT and DeiT-III [GitHub] *Mar. 2024*

- Reproduction of ViT training on ImageNet-1k and ImageNet-21k with DeiT and DeiT-III recipes.
- Trained on a TPU v4-64 pod slice, supported by the TPU Research Cloud program.

Jax/Flax Inference Codes for LLMs on TPU [GPT-NeoX] [StarCoder] *Jun. 2023*

- Implementation of GPT-NeoX (Polyglot) and StarCoder (GPTBigCode) models using Jax/Flax.
- Applied Tensor Parallelism to facilitate up to 30B models on a TPU-v3-8 node.

Korean GPT-2 Pre-Training [GitHub] [Demo] *Sep. 2020*

- A pure PyTorch implementation of GPT-2 for simple, easy and comprehensible code.
- Trained 300M model with 30GB korean corpus (5B tokens) on 2×V100 GPUs for 24 days.

COMPETITIONS

Kaggle Competitions
Kaggle Master (highest rank of 54th)

- Feedback Prize - Evaluating Student Writing [Overview] [GitHub] *Mar. 2022*
  ***Gold Medal** (4th place / 2,058 teams)*
- Google - American Sign Language Fingerspelling Recognition [Overview] [GitHub] *Aug. 2023*
  ***Gold Medal** (5th place / 1,315 teams)*
- G2Net Detecting Continuous Gravitational Waves [Overview] [GitHub] *Jan. 2023*
  Silver Medal (12th place / 936 teams)
- Learning Equality - Curriculum Recommendations [Overview] *Mar. 2023*
  Silver Medal (13th place / 1,057 teams)
- Benetech - Making Graphs Accessible [Overview] *Jun. 2023*
  Silver Medal (26th place / 608 teams)
- Google AI4Code – Understand Code in Python Notebooks [Overview] [GitHub] *Nov. 2022*
  Silver Medal (39th place / 1,135 teams)
- Feedback Prize - Predicting Effective Arguments [Overview] [GitHub] *Aug. 2022*
  Silver Medal (40th place / 1,557 teams)
- CommonLit Readability Prize [Overview] [GitHub] *Aug. 2021*
  Silver Medal (42nd place / 3,633 teams)
- Bristol-Myers Squibb – Molecular Translation [Overview] [GitHub] *Jun. 2021*
  Silver Medal (50th place / 874 teams)
- Riiid Answer Correctness Prediction [Overview] [GitHub] *Jan. 2021*
  Silver Medal (78th place / 3,406 teams)
- CSIRO - Image2Biomass Prediction [Overview] *Jan. 2026*
  Silver Medal (107th place / 3,805 teams)
- Humpback Whale Identification [Overview] *Mar. 2019*
  Bronze Medal (189th place / 2,120 teams)
- Quick, Draw! Doodle Recognition Challenge [Overview] *Dec. 2018*
  Bronze Medal (74th place / 1,309 teams)

AWARDS

Academic Awards

- **Dean's List** in the 5th semester of B.S. *Oct. 2022*
- **Dean's List** in the 3rd semester of B.S. *Oct. 2021*

ACADEMIC REVIEWING

2026: NeurIPS, ARR May, ARR Jan*
2025: ICLR, ARR Jan*, ICCV*
* Served as a secondary reviewer.